

Statistical and Machine Learning — Revision Class

Student Version

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Instructions. Work in your group. Discuss each question before writing anything down. We will review answers together after each block. You do not need to answer questions in order — prioritise the ones your group finds most interesting or challenging.

1 Block A: Linear Models

1.1 True / False

Circle **T** (true) or **F** (false) for each statement. Justify your answers.

#	Statement	T / F
1	OLS minimises the sum of squared residuals $\ \mathbf{Y} - \mathbf{X}\beta\ ^2$.	T ; F
2	The OLS estimator $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is unbiased even when the true model is non-linear.	T ; F
3	The hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is idempotent: $\mathbf{H}^2 = \mathbf{H}$.	T ; F
4	Raw residuals $\mathbf{e} = \mathbf{Y} - \hat{\beta}$ have $\mathbb{E}(\mathbf{e}) = \mathbf{0}$ always, regardless of the true model.	T ; F
5	An observation with high leverage h_{ii} is necessarily influential on the fitted values.	T ; F
6	If $\hat{\beta}_1 = 1.6$ in a logistic regression, the odds of the outcome multiply by $e^{1.6} \approx 5.0$ for a unit increase in x_1 .	T ; F
7	LDA and logistic regression always produce the same decision boundary.	T ; F

1.2 Short Conceptual Questions

Q1. Write down the logit link $g(\pi)$ and inverse-logit $h(\eta)$. Sketch (roughly) what $h(\eta)$ looks like as η ranges from $-\infty$ to $+\infty$.

Q2. Explain in one paragraph why the statement “*OLS only works when the true model is linear*” is too strong. What does OLS actually target when the true model is non-linear?

Q3. A practitioner finds Cook’s distance $D_i = 0.8$ for one observation. What does D_i measure, and what practical action would you recommend?

1.3 Worked Example

Q4. Consider a model with $n = 9$ observations and a factor variable g_i with three levels (three observations per level), along with two numerical predictors w_i and z_i :

$$Y_i = \beta_0 + \mu_{g_i} + \beta_1 w_i + \beta_2 z_i + \beta_3 z_i^2 + \beta_4 w_i z_i + \varepsilon_i.$$

- (a) How many parameters need to be estimated (assuming one level is used as a reference)?

 - (b) Is the model linear in the parameters? Is it linear in the predictors? Explain the distinction.

 - (c) Is $\hat{\beta}$ normally distributed? Under what assumption, and why?
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2 Block B: Model Evaluation

2.1 True / False

Circle **T** (true) or **F** (false) for each statement. Justify your answers.

#	Statement	T / F
8	Accuracy is always the most appropriate metric for a binary classifier.	T ; F
9	In K -fold CV, each observation appears in the validation set exactly once.	T ; F
10	Leave-one-out CV ($K = n$) always has lower variance than 5-fold CV.	T ; F
11	In a U-shaped test-error curve, the minimum occurs where training error is also minimised.	T ; F
12	The F1 score is the arithmetic mean of precision and recall.	T ; F
13	In the double-descent phenomenon, test error can decrease again even after passing the interpolation threshold $p \approx n$.	T ; F
14	Bootstrap resampling draws observations without replacement.	T ; F

2.2 Short Conceptual Questions

Q5. A binary classifier is evaluated on 200 patients (cancer screening).
TP = 40, FP = 10, FN = 20, TN = 130.

(a) Compute accuracy, sensitivity, specificity, precision.

(b) A second classifier has TP = 55, FP = 30, FN = 5, TN = 110. Compute its accuracy. Which classifier would you prefer for cancer screening, and why?

2.3 Worked Example

Q7. In K -fold cross-validation:

- (a) Why does larger K generally reduce the bias of the CV error estimate?

- (b) Give one practical reason not to always use $K = n$ (leave-one-out CV).

- (c) Describe the **double descent** phenomenon. Sketch the shape of the test-error curve as model complexity increases from $p \ll n$ to $p \gg n$. Label the interpolation threshold.

3 Block C: Regularisation and Bayesian Methods

3.1 True / False

Circle **T** (true) or **F** (false) for each statement. Justify your answers.

#	Statement	T / F
15	Ridge regression shrinks all coefficients by the same multiplicative factor for orthonormal predictors.	T ; F
16	The Lasso can set coefficients exactly to zero; Ridge cannot.	T ; F
17	Adding $\lambda \mathbf{I}_p$ to $\mathbf{X}^\top \mathbf{X}$ in Ridge regression guarantees invertibility for any $\lambda > 0$.	T ; F
18	As $\lambda \rightarrow \infty$ in Ridge regression, $\hat{\beta}_{PLS} \rightarrow \hat{\beta}_{LS}$.	T ; F
19	In the Bayesian framework, the prior is updated by the likelihood to produce the posterior.	T ; F
20	A conjugate prior always produces a posterior from the same parametric family as the prior.	T ; F
21	The MAP (maximum a posteriori) estimate under a Gaussian prior on β is equivalent to the Ridge estimator.	T ; F

3.2 Short Conceptual Questions

Q8. In the constraint-region picture for Ridge vs. Lasso, the Ridge region is a **circle** and the Lasso region is a **diamond**. Explain geometrically why Lasso tends to produce sparse solutions (exactly zero coefficients) while Ridge does not.

Q9. Write down Bayes' theorem in the form used for Bayesian inference. Name each term. Why is the normalising constant $p(\mathbf{y})$ typically omitted in practice?

3.3 Worked Example

Q10. In the penalised least squares problem:

$$PLS(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \cdot \text{pen}(\boldsymbol{\beta}).$$

- (a) Fill in the table below for orthonormal predictors ($\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$):

Method	$\hat{\beta}_j(\lambda)$	Operation on $\hat{\beta}_{j,LS}$
Ridge		
Lasso		

- (b) The MSE of the Ridge estimator decomposes as $\text{Bias}^2 + \text{Variance}$. As λ increases from 0, describe how each term changes. What shape does the MSE take, and what does this imply for choosing λ ?

- (c) Under a Gaussian prior $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_p)$ and likelihood $\mathbf{y} \mid \boldsymbol{\beta} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$, the posterior mean equals the Ridge estimator. What is the correspondence between λ and the model parameters σ^2, τ^2 ?

4 Block D: Kernel and Exemplar Methods

4.1 True / False

Circle **T** (true) or **F** (false) for each statement. Justify your answers.

#	Statement	T / F
22	In k -NN, smaller k leads to lower bias and higher variance.	T ; F
23	The kernel density estimator with bandwidth $h \rightarrow 0$ converges to a sum of Dirac deltas.	T ; F
24	A larger bandwidth h in KDE always reduces bias.	T ; F
25	The RBF (Gaussian) kernel implicitly maps data into a finite-dimensional feature space.	T ; F
26	Support vectors are the training observations closest to the decision hyperplane.	T ; F

4.2 Short Conceptual Questions

Q11. k -NN with $k = 5$ is applied to a classification problem. The five nearest neighbours of a query point have labels $(+, +, -, +, -)$.

(a) Compute $\hat{P}(Y = + | \mathbf{x})$ and state the predicted class.

(b) The same five neighbours have continuous responses $y = (3.2, 4.1, 1.0, 3.8, 2.5)$. Compute the k -NN regression prediction $\hat{f}(\mathbf{x})$.

Q12. Explain the **kernel trick** in SVMs. Why does it allow non-linear boundaries without ever computing the feature map ϕ explicitly?

4.3 Worked Example

Q13. Consider the three properties a function K must satisfy to be a valid kernel density function. Verify that the Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}}e^{-u^2/2}$ satisfies all three.

Q14. In a Gaussian Process regression, what is the effect of:

- (a) Increasing the noise variance σ^2 in the kernel?
- (b) Using a very small length-scale (large `sigma` in `rbfdot`)?

5 Block E: Tree-Based Methods and Boosting

Circle **T** (true) or **F** (false) for each statement. Justify your answers.

5.1 True / False

#	Statement	T / F
28	CART selects splits by minimising RSS (regression) or Gini impurity (classification).	T ; F
29	Bagging trains each tree on a different bootstrap sample of the data.	T ; F
30	In a bootstrap sample of size n , approximately 63% of original observations appear at least once.	T ; F
31	Random forests use all p predictors at each split, whereas bagging uses a random subset.	T ; F
32	In boosting, successive trees are fitted to the residuals of the current ensemble.	T ; F
33	In AdaBoost, a classifier with error rate $\text{err}_m = 0.5$ receives weight $\alpha_m = 0$.	T ; F

5.2 Short Conceptual Questions

Q15. A decision tree is fitted to predict salary using **Years** and **Hits**, with three terminal regions:

$$R_1 : \text{Years} < 4.5, \quad R_2 : \text{Years} \geq 4.5 \cap \text{Hits} < 117.5, \quad R_3 : \text{Years} \geq 4.5 \cap \text{Hits} \geq 117.5.$$

- (a) Identify the root node and one internal node.
- (b) A new player has Years = 6, Hits = 130. Which region does they fall into? If $\hat{c}_{R_3} = 6.74$ (mean log-salary), what is the predicted salary?

Q16. Explain in one paragraph why **random forests** have lower variance than standard bagging. Use the formula

$$\text{Var}\left(\frac{1}{M} \sum_m X_m\right) = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2.$$

5.3 Worked Example

Q17. In boosting, the weight assigned to classifier m in AdaBoost is:

$$\alpha_m = \log\left(\frac{1 - \text{err}_m}{\text{err}_m}\right).$$

Compute $\alpha_1, \alpha_2, \alpha_3$ for $\text{err}_1 = 0.1$, $\text{err}_2 = 0.3$, $\text{err}_3 = 0.45$. Interpret the results.

Q18. Explain the analogy between **gradient boosting** and gradient descent. What are pseudo-residuals? Write down their formula.

6 Open Synthesis Questions

These questions connect ideas across lectures. Discuss with your group.

S1. k -NN, KDE, and kernel regression all share the idea of local averaging. Compare how the “bandwidth” or “neighbourhood size” plays the same conceptual role in all three methods, and describe the bias-variance consequences of choosing it too small or too large.

S2. You are tasked with building a model to predict credit default (rare event: $\sim 2\%$ positive rate). Discuss: (i) which evaluation metrics you would use and why; (ii) whether you would consider regularisation; (iii) whether a tree-based ensemble might outperform logistic regression, and how you would decide.

End of student sheet.